

Cloud Security Investigation: Credential Theft, Privilege Escalation, and Data Exfiltration

This investigation focused on analyzing cloud audit logs to reconstruct a multi-stage compromise involving stolen AWS credentials, privilege escalation, persistence establishment, and sensitive data exfiltration. The objective was to identify:

1. The compromised access key
2. The attacker's infrastructure
3. Persistence mechanisms
4. Evidence of credential theft
5. Administrative privilege escalation
6. Data exfiltration activity
7. Attacker cleanup actions

Investigation Summary

The analysis revealed that an attacker obtained valid AWS credentials and used them to access cloud resources. After gaining access, the attacker:

1. Performed actions from an external IP address.
2. Established persistence using a secondary infrastructure IP.
3. Escalated privileges to gain administrative access.
4. Accessed sensitive data stored in Amazon S3.
5. Created a backdoor account for future access.
6. Disabled the compromised access key to hinder investigation.

Compromised Credentials

The stolen AWS Access Key used throughout the attack was: AKIA_DEV_PROD_9982
This access key served as the initial entry point into the cloud environment.

Attacker Infrastructure

Initial Attacker IP Address: 190.45.112.3
Secondary Persistence IP Address: 45.33.10.11

The attacker later transitioned activity to a second IP address, likely to establish persistence and maintain long-term access.

First Observed Activity From the Secondary IP: 2026-04-20T03:35:44Z

Encoded Data Analysis

During the investigation, two Base64-encoded strings were identified.

Legitimate Administrative Configuration

Encoded: c2V0dGluZzogYmFja3VwX2VuYWJsZWQ9dHJ1ZQ==
Decoded: setting: backup_enabled=true

Sensitive Credential Disclosure

Encoded: ZGJfcGFzczogUEBzc3cwcmRfQ2xvdWRfMjAyNiE=
Decoded: db_pass: P@ssw0rd_Cloud_2026!

This string contains database credentials and represents evidence of credential exposure within the environment.

Privilege Escalation

The attacker elevated their privileges by attaching an administrative policy.

Timestamp

2026-04-20T02:50:33Z

API Call

PutUserPolicy

This action granted the attacker elevated permissions and enabled unrestricted access to cloud resources.

Data Exfiltration Activity

The investigation identified unauthorized access to sensitive data stored in Amazon S3.

Target Bucket

top-secret-financials-2026

Downloaded File

ceo_personal_taxes_2025.pdf

The file was successfully accessed and downloaded by the attacker, confirming data exfiltration.

Persistence Mechanism

To maintain long-term access, the attacker created a new user account.

Backdoor User

support_service_backup

This account provided an alternative access path even if the original credentials were detected and revoked.

Anti-Forensics and Cleanup

After completing their objectives, the attacker took steps to reduce the likelihood of detection.

Action Performed

UpdateAccessKey

Result

Status = Inactive

The attacker disabled the originally compromised access key after establishing persistence and exfiltrating data.

This tactic can complicate investigations by making the compromised credential appear inactive during later reviews.

Skills Demonstrated

1. Cloud Security Monitoring
2. AWS Log Analysis
3. Incident Response
4. Threat Hunting
5. Credential Theft Investigation
6. Privilege Escalation Analysis
7. Data Exfiltration Detection
8. Cloud Forensics
9. MITRE ATT&CK Mapping